

CYBERSECURITY

LAB 5

1. Introduction

The laboratory aims to become familiar with elements of cryptanalysis on symmetrical and asymmetrical algorithms. Below is a list of terms containing theoretical attack models and their practical application.

Bruteforce attacks

This type of attack consists of checking every possible combination in a key to find the "key" to correctly decipher the message. Bruteforce attacks can take less time for smaller keyspaces, but for larger key spaces they take an extremely long time. Bruteforce attacks are unpractical for modern encryption algorithms.

Ciphertext only attacks

It is also called "known ciphertext attacks". In a "cipher-only attack", the attacker knows the ciphertext of various messages that have been encrypted using the same encryption algorithm. The attacker must find a key which can then be used to decrypt all messages. This is one of the easiest attacks to carry out because it is easy to intercept the ciphertext (by sniffing), but it is difficult to implement because knowledge of the encryption process is limited. The methods that can be used:

- Attack on Two-Time Pad,
- Frequency Analysis,

Chosen-Plaintext Attack

A "chosen-plaintext" attack is similar to a 'known plaintext' attack, but in this case, the attacker selects their texts to help them crack the key. A properly selected sequence of characters with generated cipher can restore the key. The found key can be used to obtain information about the whole encryption process and understand how it is carried out. The attacker analyses the system behavior and output ciphertext, based on any kind of input data.

Known-plaintext Attack

In a known-plaintext attack, the attacker knows some of them and encrypts them. Reverse engineering is then used to find the key. Attacks of this type were only used to break simple ciphers.

Differential cryptanalysis

The attacker observes changes in two texts up to an encrypted form. Based on the changes taking place, the attacker tries to find the key. This is a kind of 'chosen-plaintext' attack because the attacker selects plain text to observe the changes. The attack was used to break symmetrical algorithms like DES, but over time, the subsequent algorithms have become resistant.

Linear cryptanalysis

An attacker carries out a known-plaintext attack on several messages that have been encrypted with the same key. This gives the attacker an insight into the probability of a particular key occurring. If more than one message is attacked, there is a higher probability of finding a specific "key".

Man-in-the-Middle Attack

The attacker joins the communication as a hidden intruder. With two keys, he intercepts communication between the two entities. Using the previously prepared set of keys, the attacker decrypts, intercepts the content, encrypts, and then sends it to the target entity. To minimize the occurrence of such an attack, strong authentication methods must be used. An example is the use of asymmetric algorithms or key exchange using the Diffie-Hellman protocol.

Frequency analysis

Frequency analysis is similar to the attack on a known ciphertext. The attack consists of analyzing letters and groups of letters for a specific language. Knowledge of a language is important because each language has its frequency of occurrence of given letters. Popular characters in the ciphertext are searched for and converted into popular characters in a given language. Substitution Ciphers group can be given this type of attack.

Replay attack

An intruder sends the same message to the victim as in the victim's communication impersonating one of the parties. This allows the intruder to be authenticated by one of the parties and treated as a trusted entity in the communication. The intruder may gain unauthorized access and may extract the desired information. This type of attack is similar to a Man-in-the-Middle attack.

2. Prerequisites

In this laboratory, the Cryptool tool is used to perform some process of cryptanalysis. The materials available at the lecture should be read. In Cryptool, options from the Analyses tab is used. Get acquainted with the cryptanalysis elements (types).

3. Cryptanalysis for the symmetric algorithm**I. Tasks:**

1. Evaluate the time needed to find a full key of 64, 128, 192, 256 bits.
2. Compare the time it takes to find keys of the same length (e.g. 128 bits) for different algorithms.
3. Determine the time needed to find a key if the unknown number of bits of the key is: 4, 8, 12, 16, 20, 24, 30, 34. (One star in the key equals 4 bits)
4. Check whether the position of the unknown bits in the key affects the key search time.
5. Evaluate the performance of the cryptanalysis tools.
6. Do we always get the correct key?
7. Does the number of the bits search affect the quality of the key being restored?
8. Does the position of unknown bits affect the quality of the key being

restored?

II. Questions:

1. Can modern block algorithms be considered safe (referring to the above experiments)?
2. What length of the key offers us a sufficient level of security for particular algorithms? Why?
3. Does the size of the ciphertext influence the possibility of breaking it?
4. Does the format and earlier processing of the document affect the possibility of its cryptanalysis (e.g. compression etc.)?
5. How many possible passwords can we check in a year of continuous work on one computer, which checks one million passwords per second ($\approx 2^{20}$)?
6. What does this result say about the security of modern algorithms?

4. Cryptanalysis for the asymmetric algorithm

I. Tasks:

1. Please try to factorize the number created by the combination of
 - Student's ID number
 - Year
 - Month
 - Day
 - Hour
 - MinutesExample: 123456202011051408

Tab: Indiv. Procedures -> RSA Cryptosystem -> Factorization of a number

2. Please check how the time of searching for prime numbers increases with the value of searching interval:

Tab: Analysis -> Asymmetric encryption -> Generate Prime Numbers

3. Check how different parameter values influence the effectiveness and time of need for the RSA module N factorization attack:

Length N, Length p, Length of known bit string P (lab_5_numbers.txt)

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Tab: Analysis -> Asymmetric encryption -> Lattice-Based Attacks on RSA -> Factoring with a hint

4. Find the missing text fragment from lab_5_number.txt using the attack method available in the tab:

Tab: Analysis -> Asymmetric encryption -> Lattice-Based Attacks on RSA -> Attack on Stereotyped Messages

5. Factorize the modules from the file lab_5_number.txt using the additional data contained in this file (values: d, e, delta, ...) and the attack method available under the tab:

Tab: Analysis -> Asymmetric encryption -> Lattice-Based Attacks on RSA -> Attack on Small Secret Keys

II. Questions:

1. What is the minimum length of the module (number N) of the RSA algorithm, which guarantees that its distribution (finding its prime factors) will be difficult enough?
2. Is it possible to carry out an effective factoring attack based on partial knowledge of the value of one parameter for the module length assumed in the previous point as safe? (task 3)?
3. In what cases can RSA encryption be threatened by the attack in point 4?
4. In what cases can RSA encryption be threatened by the attack in point 5?